Figure 0B: Synteny-Anchored Locus Retrieval Pipeline & 4 Archetypes

Synteny Discovery, HMMER Scanning & Chromosome Validation Pipeline

1. Syntenic Flank Search

Query 68 genomes for anchor markers:
α: NPRL3, MPG, RHBDF1, LUC7L
β: OR51 cluster, TRIM21, RRM1

2. HMMER / TBLASTN Scan

When annotations missing/divergent:
Profile HMM scan (Pfam PF00042)
Cross-species seeded TBLASTN

3. De Novo Gene Models

Extract canonical 3-exon / 2-intron globin coding sequences (GT-AG); validate invariant heme pocket

4. Scaffold vs. Chromosome

Discriminate fully assembled arms from unplaced contigs to guarantee true telomere distance measurement

5. Baseline Calibration

Compute whole-genome CDS baselines (N = 15k–35k genes) to quantify isochore z-score deviations (ΔGC3)

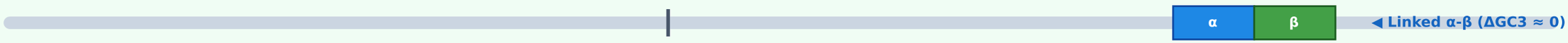
The 4 Core Macroevolutionary Archetypes Across Vertebrate Evolution

1. Linked Primordial Archetype (Basal Actinopterygii & Amphibians — Spotted Gar, Xenopus)

Ancestral tandem arrangement: α and β globin genes reside on the same chromosome in a single cluster.

Both genes share identical genomic environment and regional recombination rate.ΔGC3 (α - β) ≈ 0%.

Cytogenetic State: Single Chromosome (Gar Chr 19 / Xenopus Chr 1)



2. Canonical Decoupled Subtelomeric-α (Eutherian & Metatherian Mammals — Human, Mouse)

Amniote translocation decouples α and β. α captured by subtelomere (<1 Mb to telomere); β remains interstitial.

Runaway subtelomeric gBGC supercharges α-GC3 (>85–95%) while β-GC3 remains moderate (~60–65%). ΔGC3 > 0%.

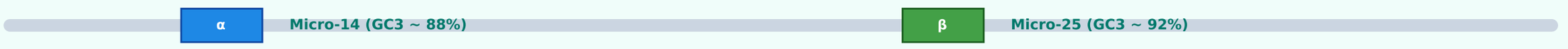
Cytogenetic State: Two Chromosomes (α Subtelomeric / β Interstitial)

3. Dual Microchromosomal Partitioning (Archosauria & Aves — Chicken, Eagle, Hummingbird)

Both α and β clusters translocate onto distinct, highly recombinogenic microchromosomes (Chr 14 and Chr 25).

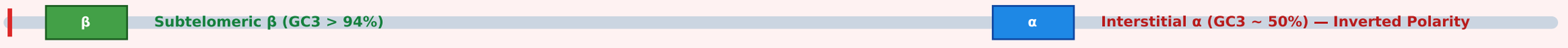
High crossover density drives synchronous hyper-elevation of both clusters:α-GC3 > 85%, β-GC3 > 85%. ΔGC3 ≈ 0%.

Cytogenetic State: Two Microchromosomes (Micro-14 and Micro-25)



4. Squamate Inverted Polarity (Lizards & Snakes — Wall Lizard, Garter Snake)

Squamates exhibit inverted chromosomal polarity: the β locus is translocated to a microchromosome or subtelomere,

driving elevated β-GC3 (85–95%), while the α locus resides in an interstitial genomic desert (GC3 48–55%). ΔGC3 < 0%.

Cytogenetic State: Inverted Polarity (β Subtelomeric / α Interstitial)